

Darshan Singh S

Pre-Doctoral Researcher, Google DeepMind

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Education

International Institute of Information Technology Hyderabad (IIIT-H) 2022–2024
M.S by Research in Computer Science CGPA: 9.67/10.0

Experience

Google DeepMind Nov 2024 - Present
Pre-Doctoral Researcher

Projects: Gemini Core team member - multimodal-multicultural video understanding and reasoning, multimodal pretraining and posttraining evals, knowledge probing, cross-lingual gaps analysis in language models.
Languages Team: I primarily work with [Partha Talukdar](#), [Shachi Dave](#), [Arsha Nagrani](#) and [Tobias Weyand](#)

FastCode AI June 2024 - Nov 2024
Machine Learning Engineer

Projects: Developed a time-series foundation model for the electric vehicles data for Mercedes-Benz Research & Development, fine-tuned stable diffusion models and adapted them to video games.
Team: [Dr. Arjun Jain](#), Mercedes-Benz Research & Development, India

CVIT Lab, IIIT Hyderabad July 2021 - Aug 2022
Research Fellow | Advisors: Prof. C. V. Jawahar and Prof. Makarand Tapaswi

Projects: Audio-Visual lecture segmentation, published at WACV 2023.
Team: Video-Language (Multimodal) team

Mercedes Benz R&D India Aug 2019 - Aug 2020
Graduate Engineer Trainee

Projects: Implemented graph algorithms for enhancing functionalities of a CAD software platform called Micro-Station.

Publications

†=Equal Contribution

- [1] **VELOCITI: Benchmarking Video-Language Compositional Reasoning with Strict Entailment** 📄 🤖
Darshana S[†], Varun Gupta[†], **Darshan Singh S[†]**, Zeeshan Khan, Vineeth Gandhi, Makarand Tapaswi.
Computer Vision and Pattern Recognition (CVPR). [CVPR'25]
- [2] **CURVE: A Benchmark for Cultural and Multilingual Long Video Reasoning.** 📄
Darshan Singh, Arsha Nagrani, Kawshik Manikantan, Harman Singh, Dinesh Tewari, Tobias Weyand, Cordelia Schmid, Anelia Angelova, Shachi Dave. *Under Review, 2025*.
- [3] **FiGCLIP: Fine Grained CLIP Adaptation via Densely Annotated Videos** 📄
Darshan Singh S, Zeeshan Khan, Makarand Tapaswi. *arXiv Preprint, 2024*.
- [4] **Unsupervised Audio-Visual Lecture Segmentation** 📄
Darshan Singh S[†], Anchit Gupta[†], C. V. Jawahar, Makarand Tapaswi.
Winter Conference on Applications of Computer Vision (WACV). [WACV'23]
- [5] **No Detail Left Behind: Revisiting Self-Retrieval for Fine-Grained Image Captioning** 📄
Manu Gaur, **Darshan Singh S**, Makarand Tapaswi.
Transactions of Machine Learning Research (TMLR). [TMLR'24]
🏆 Top 10% Distinction, to be presented at ICLR 2026.
- [6] **D3: Evaluating MLLMs capacity for Finegrained Visual Discrimination through Self-Retrieval** 📄
Manu Gaur, **Darshan Singh S**, Makarand Tapaswi.
ECCV-Eval-FoMo Workshop. [ECCV-W'24]
- [7] **Rethinking Cross-lingual Gaps from a Statistical Viewpoint** 📄
Vihari Piratla, Purvam Jain, **Darshan Singh S**, Trevor Cohn, Partha Talukdar. *Under Review, 2025*.

Selected Projects

Gemini (Core-team member)

Google DeepMind

Team: Partha Talukdar, Arsha Nagrani, Shachi Dave, Tobias Weyand, Anelia Angelova, Cordelia Schmid

- › Developed multimodal-multicultural video understanding and reasoning benchmarks which are used in pre-training and post-training evals.
- › Led an effort on reliable knowledge estimation in LLMs and contributed to quantifying cross-lingual gaps in LLMs.

FiGCLIP: Fine-Grained CLIP Adaptation via Densely Annotated Videos

IIIT Hyderabad

Advisors: Prof. Makarand Tapaswi

- › Addressed the lack of fine-grained and syntactic information in CLIPs representations by adapting CLIP on holistic, multidimensional, and densely annotated video-text data.
- › Proposed a lightweight adaptation strategy with LoRA adapters that enable learning fine-grained representations without catastrophic forgetting.
- › Outperformed the base CLIP across 5 diverse tasks including video situation recognition, zero-shot action recognition, dense video captioning, and VL compositional reasoning.

VELOCITI: Benchmarking Video-Language Compositional Reasoning

IIIT Hyderabad

Advisors: Prof. Makarand Tapaswi, Vineeth Gandhi

- › Proposed VELOCITI, a new benchmark to evaluate how well Video-LLMs understand compositional reasoning, and introduced StrictVLE, a stricter evaluation method.
- › Key finding: State-of-the-art models like Gemini 1.5 Pro (49.3%) perform far below human accuracy (93.0%), showing significant struggles in associating agents with actions.
- › Provided key insights, such as understanding actions is harder than agents for open models and smaller models are predisposed towards Yes for the entailment task.

No Detail Left Behind: Revisiting Self-Retrieval for Fine-Grained Image Captioning

Advisors: Prof. Makarand Tapaswi

- › We design- (1) a post-training recipe for self-retrieval finetuning with REINFORCE, and (2) a synthetic framework for visually boosting captioning datasets
- › Jointly, they enable captioners to generate fine-grained, succinct descriptions while reducing hallucinations.
- › Using our training recipe, ClipCap, a 200M param simplification of modern MLLMs, outperforms sota open-source MLLMs on fine-grained visual discrimination.

Unsupervised Audio-Visual Lecture Segmentation

IIIT Hyderabad

Advisors: Prof. C. V. Jawahar, Prof. Makarand Tapaswi

- › Proposed a method to segment video lectures into bite-sized topics using visual, textual, and OCR cues via a self-supervised pretext task.
- › Utilized learned representations to temporally segment lectures using the TW-FINCH algorithm.
- › Introduced AVLectures, a new large-scale dataset of 86 courses (2,350+ lectures) from MIT-OpenCourseWare for pre-training, fine-tuning, and evaluation.

Technical Skills

Languages	Python, C, C++, MATLAB, Assembly Language(8051), Embedded C
Tools/Libraries	PyTorch, NumPy, scikit-learn, L ^A T _E X, Git, HuggingFace

Relevant Coursework

Statistical Methods in AI*, Topics in Deep Learning*, Advanced NLP, Computer Vision, Data Structures and Algorithms*, Digital Image Processing*, Fairness, Privacy & Ethics in AI*, Operating Systems, Programming in C*, Advanced Calculus*, Multivariable Calculus*, Probability and Statistics.

(* indicates grade A for outstanding performance)

Selected Honors, Awards and Academic Service

- › Awarded J2C Certification (top 10%) for our paper at TMLR 2024
- › Selected for competitive Pre-doctoral position at Google DeepMind
- › Presented my work on reliable knowledge estimation at Google DeepMind APAC forum and Multilinguality Research Forum

- › Gave a talk on Computer Vision Fundamentals along with Pytorch tutorials at the CVIT Computer Vision Summer School, IIIT Hyderabad
- › Started Vision Reading Group at IIIT Hyderabad
- › Reviewer: ECCV 2024 (including emergency reviewing), ACCV 2025, CVPR 2025 (including emergency reviewing)

References

References from the following available upon request:

- › Partha Talukdar Principal Scientist, Google DeepMind 
- › Arsha Nagrani Senior Research Scientist, Google DeepMind 
- › Anelia Angelova Principal Scientist, Google DeepMind 
- › Shachi Dave Research Engineer, Google DeepMind 
- › Prof. C.V. Jawahar Professor, Dean R&D, IIIT Hyderabad 
- › Prof. Makarand Tapaswi Assistant Professor, IIIT Hyderabad 
- › Prof. Vineet Gandhi Associate Professor, IIIT Hyderabad 